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CC - Assignment 02

(Q.1) Concept of Virtualization (Types, Structure, Mechanism)

- 1] It is process of creating virtual versions of physical resource like servers, storage or networks.
- 2] A hypervisor manages virtualization by allocating CPU memory to each VM, Each VM runs its own OS independently.

Types : Hardware Virtualization  
OS level Virtualization  
Storage Virtualization  
Network Virtualization.

Structures :

Type 1 (Base-Metal) : Runs directly on hardware.

Type 2 (Hosted) : Runs on top of OS, easier to use.

Mechanisms : Full virtualization : No OS modification needed.

Para Virtualization : OS modified for efficiency

Hardware assisted : Uses CPU support for better performance.

Q2] Virtual box vs KVM.

| → Parameter      | Virtual Box                           | KVM.                                   |
|------------------|---------------------------------------|----------------------------------------|
| Type             | Type 2 (Hosted Hypervisor)            | Type 1 (Base-monitor)                  |
| Architecture     | Run on top of host OS.                | Integrated directly into Linux kernel. |
| Performance      | Moderate                              | High.                                  |
| Ease of use      | Very easy                             | Moderate to difficult.                 |
| Scalability      | Limited (personal use)                | Highly Scalable                        |
| Security         | Basic Isolation                       | Strong                                 |
| Resource usage   | Less efficient                        | Highly efficient                       |
| Hardware Support | Limited                               | Full hardware.                         |
| Portability      | High (Runs on windows, Linux, macOS)  | Linux - Only                           |
| Use case         | Testing, development Cloud Computing. |                                        |



Q.4 } Load Balancing, Auto Scaling, Security in VMs

→ Load Balancing:

Distributes workload across multiple VMs to avoid overload.

- Improves performance.
- Ensures high availability.

Auto - Scaling:

Automatically increases or decreases VMs based on demand.

- Scale - out when traffic increases.
- Scale - in when demand drops.
- Saves cost & resources.

Security Services:

Isolation: VMs are separated.

Firewalls: Control traffic.

Encryption: protect data.

Access Control: Authentication.

Monitoring: Detect threats.